

LECTURE 05 · REVIEW NOTES

Interest Rate Parity

The no-arbitrage relationship between spot rates, forward rates, and interest rate differentials — and how it governs corporate hedging.

About these notes

These notes accompany Lecture 05 of FINA 405 · International Financial Management (University of San Diego, Knauss School of Business, Fall 2026). Content is presented in textbook style. All numerical examples are independently verified. In-class exercises are described by topic and structure only.

1. FX Exposure: Assets, Liabilities, and Risk Direction

A firm that has future cash flows denominated in a foreign currency faces FX risk. The direction of that risk depends on whether the foreign currency position is an asset (receivable) or a liability (payable).

Position	Type	FX Risk Arises When...	Impact on Firm
Foreign Currency Receivable (e.g., export proceeds)	Asset	Foreign currency depreciates vs. USD	Fewer USD received upon conversion — revenue falls short of plan.
Foreign Currency Payable (e.g., import invoice)	Liability	Foreign currency appreciates vs. USD	More USD required to settle the obligation — cost exceeds budget.

IN-CLASS ACTIVITY · FX Risk Quick Think

At the start of class, students identified the direction of FX risk for two scenarios: (A) a U.S. firm that will receive €1 million in six months (foreign-currency asset), and (B) a U.S. firm that must pay €1 million in six months (foreign-currency liability). For scenario A, EUR depreciation reduces the USD value of the receivable. For scenario B, EUR appreciation increases the USD cost of the payable. This framing motivates the core question of the lecture: how can the firm eliminate this uncertainty by locking in a rate today?

2. The Importer's Problem

Consider a U.S. firm that must pay €1,000,000 to a German supplier in six months. The current spot rate is EUR/USD = 1.0800. Two outcomes are possible:

Scenario	EUR/USD at Maturity	USD Cost	vs. Today's Plan (\$1,080,000)
EUR appreciates	1.15	\$1,150,000	+\$70,000 unexpected loss
EUR depreciates	1.02	\$1,020,000	–\$60,000 windfall saving

The solution is to buy EUR forward: the firm enters a forward contract today, agreeing to purchase €1,000,000 at a fixed rate on the due date. The fair price for that forward rate is given by Interest Rate Parity.

3. Interest Rate Parity: Concept and Formula

Interest Rate Parity (IRP) states that the forward exchange rate must reflect the interest rate differential between two currencies. If it did not, a riskless profit would be available through covered interest arbitrage, and arbitrageurs would immediately trade the gap away. The no-arbitrage condition yields:

$$F = S \times (1 + r_{\text{quote}} \times T) / (1 + r_{\text{base}} \times T)$$

Symbol	Definition
F	Forward exchange rate (quote-currency units per one unit of base currency)
S	Spot exchange rate (today)
r_quote	Interest rate in the quote currency (domestic, e.g., USD rate for EUR/USD)
r_base	Interest rate in the base currency (foreign, e.g., EUR rate for EUR/USD)
T	Maturity in years (3M = 0.25, 6M = 0.50, 12M = 1.00)

Intuition: a currency with a higher interest rate must depreciate in the forward market by just enough to offset the interest advantage. If $r_{\text{quote}} > r_{\text{base}}$, then $F > S$ (the quote currency trades at a forward premium). If $r_{\text{quote}} < r_{\text{base}}$, then $F < S$ (the quote currency trades at a forward discount).

WORKED EXAMPLE 1

Deriving the No-Arbitrage Forward Rate from First Principles

Time / Step	Action	Amount
t = 0	Borrow €1,000,000 at $r_{\text{EUR}} = 3\%$ for 1 year	—
t = 0	Convert to USD at spot EUR/USD = 1.0850	\$1,085,000
t = 0	Lend \$1,085,000 in the U.S. at $r_{\text{USD}} = 5\%$ for 1 year	—
t = 1	Receive USD principal + interest: $\$1,085,000 \times 1.05$	\$1,139,250
t = 1	Repay EUR loan + interest: $\text{€}1,000,000 \times 1.03$	€1,030,000
t = 1	Implied no-arb forward: $\$1,139,250 \div \text{€}1,030,000$	$F = 1.1061$
Verify	$F = 1.0850 \times (1.05 / 1.03) = 1.0850 \times 1.019417\dots$	$= 1.1061 \checkmark$

The forward rate of 1.1061 is not a forecast — it is the only rate at which no arbitrage profit can be earned by borrowing in one currency and investing in the other.

4. IRP in Practice: High and Low Rate Environments

The IRP formula generates a full term structure of forward rates — a forward curve — for any currency pair. The steepness of that curve is driven by the interest rate differential.

Currency Pair	Spot	r_USD	r_Foreign	Differential	Forward Direction
USD/BRL (Brazil)	5.20	5.25%	14.75%	+9.50%	BRL at forward discount — more BRL per USD over time (BRL depreciates).
USD/JPY (Japan)	149.50	5.25%	0.10%	−5.15%	JPY at forward premium — fewer JPY per USD over time (JPY appreciates).

5. Covered Interest Arbitrage

When the market forward rate deviates from the IRP-implied forward rate, covered interest arbitrage (CIA) generates a riskless profit. The strategy requires no forecasting: all legs are locked in simultaneously.

CIA mechanism when $F_{\text{market}} < F_{\text{IRP}}$ (forward is too cheap): ① Borrow the foreign currency; ② Convert to domestic at spot; ③ Lend domestic at r_{domestic} ; ④ Buy the foreign currency forward at F_{market} . The discrepancy between the proceeds and the repayment obligation is the risk-free profit.

WORKED EXAMPLE 2

Spotting and Exploiting an EUR/USD Mispricing

Step	Calculation	Result
Given	EUR/USD spot = 1.0850; $r_{\text{USD}} = 5.25\%$; $r_{\text{EUR}} = 3.75\%$; $T = 1$ yr; $F_{\text{mkt}} = 1.0900$	—
Compute F_{IRP}	$F_{\text{IRP}} = 1.0850 \times (1.0525 / 1.0375)$	= 1.1007
Compare	$F_{\text{mkt}} = 1.0900$ vs. $F_{\text{IRP}} = 1.1007$	EUR underpriced forward
① Borrow EUR	€921,659 at 3.75%; repayment at $T=1$	€956,221 owed
② Convert at spot	$€921,659 \times 1.0850$	\$1,000,000 USD
③ Invest USD	\$1,000,000 at 5.25% for 1 year	\$1,052,500
④ Buy EUR fwd	$€956,221 \times F_{\text{mkt}} = €956,221 \times 1.0900$	\$1,042,281 cost
Profit	$\$1,052,500 - \$1,042,281$	+\$10,219 risk-free

The arbitrage profit arises because the market forward (1.0900) underprices EUR relative to the no-arb rate (1.1007). The CIA trade buys EUR forward at the cheap market rate, generating \$10,219 per \$1 million invested, with zero market risk.

IN-CLASS ACTIVITY · CIA Exercise

Students worked through a full covered interest arbitrage exercise using the GBP/USD pair: Spot = 1.2500, $r_{\text{USD}} = 5\%$, $r_{\text{GBP}} = 2\%$, $T = 1$ year, $F_{\text{market}} = 1.3000$. Students were asked to identify which currency was overvalued in the forward market, which to borrow, which to invest in, and to calculate the approximate arbitrage profit per \$1,000,000. The exercise was worked step by step in class and submitted individually for grading.

6. Multinational Application: Hedging a Payable

A forward hedge converts an uncertain foreign-currency obligation into a known USD cost, enabling precise financial planning. The following case illustrates the full mechanics, including the outcome under different realized spot scenarios.

WORKED EXAMPLE 3

TechGlobal Inc. — Hedging an 18-Month EUR Payable

Input / Step	Data & Calculation	Value
EUR/USD Spot (S)	Given	1.0850
r_USD (r_quote)	Given	5.25% p.a.
r_EUR (r_base)	Given	3.75% p.a.
Maturity (T)	18 months = 1.5 years	1.5 yr
F(18M) numerator	$1 + 0.0525 \times 1.5 = 1 + 0.07875$	1.07875
F(18M) denominator	$1 + 0.0375 \times 1.5 = 1 + 0.05625$	1.05625
F(18M) = IRP rate	$1.0850 \times 1.07875 / 1.05625$	= 1.1081
USD cost of hedge	$\text{€}2,500,000 \times 1.1081$	\$2,770,250

By entering a forward contract at 1.1081, TechGlobal eliminates all FX uncertainty on its €2.5M payable. The USD cost is known at inception, regardless of where spot trades at settlement.

Hedge Outcome Under Alternate Spot Scenarios

Scenario	Realized Spot	Without Hedge	With Hedge	Hedge Impact
EUR strengthened	1.1500	$\text{€}2.5\text{M} \times 1.1500 = \$2,875,000$	$\text{€}2.5\text{M} \times 1.1081 = \$2,770,250$	Saved \$104,750
EUR weakened	1.0600	$\text{€}2.5\text{M} \times 1.0600 = \$2,650,000$	$\text{€}2.5\text{M} \times 1.1081 = \$2,770,250$	Cost extra \$120,250

A forward hedge is insurance, not speculation. The firm sacrifices the potential upside (paying less if EUR weakens) in order to eliminate the downside (paying more if EUR strengthens). Crucially, a forward contract carries zero upfront cost — no premium is paid at inception, unlike an option. The cost of the hedge is embedded in the forward rate itself via the interest rate differential.